Table 1: Diagnostic fixed-parameter CatBoost analysis in five prespecified scenarios. Original logistic spy, full CatBoost nonlinear spy, size-matched nonlinear spy, random size-matched and oracle true-negative size-matched controls are compared. Random/oracle draws are first averaged within each repetition; repetitions are then equally weighted (100 per setting/design). Oracle controls use hidden truth and are not practical PU methods. Nonlinear-spy RN is not added to the benchmark catalog. Plotted absolute AUROC/AP scales are 0 to 1.

One-sample - Reference				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.726	0.395	100	100
Nonlinear spy full	0.719	0.381	100	100
Nonlinear size-matched	0.720	0.385	100	100
Random size-matched	0.715	0.389	100	100
Oracle TN size-matched	0.721	0.401	100	100

One-sample - High dimension ($p = 500$)				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.719	0.387	100	100
Nonlinear spy full	0.711	0.369	100	100
Nonlinear size-matched	0.709	0.363	100	100
Random size-matched	0.681	0.345	100	100
Oracle TN size-matched	0.698	0.367	100	100

One-sample - Curved boundary				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.695	0.348	100	100
Nonlinear spy full	0.699	0.355	100	100
Nonlinear size-matched	0.700	0.357	100	100
Random size-matched	0.696	0.362	100	100
Oracle TN size-matched	0.704	0.375	100	100

One-sample - Feature interaction				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.614	0.260	100	100
Nonlinear spy full	0.701	0.347	100	100
Nonlinear size-matched	0.696	0.338	100	100
Random size-matched	0.699	0.369	100	100
Oracle TN size-matched	0.709	0.386	100	100

One-sample - Unequal class covariance				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.688	0.370	100	100
Nonlinear spy full	0.690	0.354	100	100
Nonlinear size-matched	0.692	0.356	100	100
Random size-matched	0.717	0.438	100	100
Oracle TN size-matched	0.724	0.457	100	100

Two-sample - Reference				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.726	0.393	100	100
Nonlinear spy full	0.719	0.378	100	100
Nonlinear size-matched	0.720	0.382	100	100
Random size-matched	0.715	0.386	100	100
Oracle TN size-matched	0.723	0.403	100	100

Two-sample - High dimension ($p = 500$)				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.716	0.383	100	100
Nonlinear spy full	0.710	0.369	100	100
Nonlinear size-matched	0.707	0.362	100	100
Random size-matched	0.674	0.336	100	100
Oracle TN size-matched	0.698	0.367	100	100

Two-sample - Curved boundary				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.696	0.350	100	100
Nonlinear spy full	0.700	0.358	100	100
Nonlinear size-matched	0.701	0.360	100	100
Random size-matched	0.696	0.359	100	100
Oracle TN size-matched	0.706	0.377	100	100

Two-sample - Feature interaction				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.614	0.261	100	100
Nonlinear spy full	0.702	0.351	100	100
Nonlinear size-matched	0.699	0.344	100	100
Random size-matched	0.695	0.361	100	100
Oracle TN size-matched	0.710	0.386	100	100

Two-sample - Unequal class covariance				
Selector/control	AUROC	AP	n AUROC	n AP
Logistic spy	0.687	0.371	100	100
Nonlinear spy full	0.691	0.357	100	100
Nonlinear size-matched	0.693	0.359	100	100
Random size-matched	0.714	0.432	100	100
Oracle TN size-matched	0.724	0.458	100	100

Note. Fixed CatBoost final-refit parameters. Multiple random/oracle draws are first averaged within repetition. Absolute means are conditional on the original fixed-parameter experiment protocol; these controls are diagnostics, not additional benchmark configurations.